



Development of blood brain barrier permeation prediction models for organic and inorganic biocidal active substances

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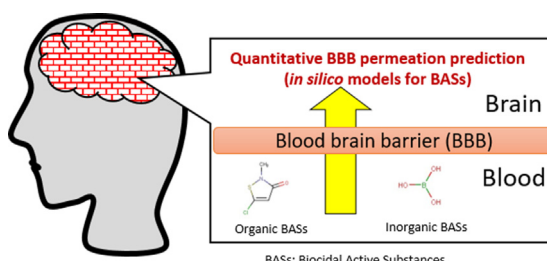
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HIGHLIGHTS

- logBB prediction models for biocides were developed.
- logBB prediction model for organic biocides outperformed that for drug-like compounds.
- logBB of inorganic compounds was significantly correlated with water solubility and pyrolysis point.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 19 January 2021

Received in revised form

16 March 2021

Accepted 16 March 2021

Available online 18 March 2021

Handling Editor: Willie Peijnenburg

Keywords:

QSAR

In silico

Safe-by-design

BBB

Computational neurotoxicology

Biocide

ABSTRACT

Biocidal products are broadly used in homes and industries. However, the safety of biocidal active substances (BASs) is not yet fully understood. In particular, the neurotoxic action of BASs needs to be studied as diverse epidemiological studies have reported associations between exposure to BASs and neural diseases. In this study, we developed *in silico* models to predict the blood-brain barrier (BBB) permeation of organic and inorganic BASs. Due to a lack of BBB data for BASs, the chemical space of BASs and BBB dataset were compared in order to select BBB data that were structurally similar to BASs. *In silico* models to predict log-scaled BBB penetration were developed using support vector regression for organic BASs and multiple linear regression for inorganic BASs. The model for organic BASs was developed with 231 compounds (training set: 153 and test set: 78) and achieved good prediction accuracy on an external test set ($R^2 = 0.64$), and the model outperformed the model for pharmaceuticals. The model for inorganic BASs was developed with 11 compounds ($R^2 = 0.51$). Applicability domain (AD) analysis of the models clarified molecular structures reliably predicted by the models. Therefore, the models developed in this study can be used for predicting BBB permeable BASs in human. These models were developed according to the Quantitative Structure-Activity Relationship validation principles proposed by the Organization for Economic Cooperation and Development.

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1. Introduction

Biocidal products (BPs) are used in homes for hygienic purposes (Smolka, 2013) or in industries to control the growth of bacteria (Collier et al., 1990; Kahrilas et al., 2015). As the usage of BPs is

Abbreviations

AD	Applicability domain
BAS	Biocidal active substance
BBB	Blood-brain barrier
BP	Biocidal product
BPR	Biocidal products regulation
CNS	Central nervous system
CV	Cross Validation
FP	Fingerprint
logBB	log-scaled blood brain barrier penetration
logBB _[inorg]	logBB data set for inorganic compounds
logBB _[org]	logBB data set for organic compounds
logS	Water solubility
MACCS FP	Molecular access system fingerprint

MIT	Methylisothiazolinone
MLR	Multiple linear regression
MMFF	Merck molecular force field
Model _[inorg]	Prediction model for logBB of inorganic compounds
Model _[org]	Prediction model for logBB of organic compounds
MOPAC	Molecular Orbital PACKage
MW	Molecular Weight
T _{BP}	Boiling point
T _{MP}	Melting point
T _{PP}	Pyrolysis point
QM	Quantum mechanics
RFR	Random forest regression
RMSE	Root mean squared error
SVR	Support vector regression

intended to have harmful effects on living organisms, BPs are potentially hazardous to human health. According to a study on consumer exposure to biocidal active substances (BASs), a large number of consumers can be exposed to BASs through a wide range of BPs used in homes (Hahn et al., 2010). It has been reported that approximately 15,600 people in Europe suffered from adverse outcomes because of BPs between 2003 and 2005 (Schweer, 2010). The humidifier disinfectant scandal in South Korea is another unfortunate incident in which BPs caused heavy casualties (Pangyi and Jong-Han, 2016). Due to the hazardous effects of BASs on human health, BPs are regulated by the biocidal products regulation (BPR) in Europe (Sobanska et al., 2003; ECHA, 2016). In Korea, K-BPR has come into force since 2019.

Neurotoxic agents damage the structure or function of the nervous system, which can lead to the development of neurodegenerative disorders (Soleimani et al., 2016). Among the various types of BAS, the neurotoxicity of insecticides has been widely studied as these chemicals affect the nervous system of insects. As the nervous system of insects and humans is significantly similar, most insecticides can damage the nervous system of humans (Costa, 2019). Epidemiological studies have suggested possible associations between exposure to BASs and neural disorders (Roberts Eric et al., 2007; London et al., 2012; Wagner-Schuman et al., 2015; Lee et al., 2020). Several *in vitro* studies have reported on the neurotoxicity of diverse BASs (Du et al., 2002; He et al., 2006; Park et al., 2016; Christen et al., 2017; Tong et al., 2017; Silva et al., 2020). However, further studies are required to identify neurotoxic BASs. In the case of methylisothiazolinone (MIT), *in vitro* studies reported that MIT was neurotoxic as it damaged neuronal cells. However, no evidence of neurotoxicity was reported in another *in vivo* study on MIT, and the lack of systemic toxicity was attributed to rapid metabolism and excretion of MIT in rats and mice (Burnett et al., 2010).

The blood-brain barrier (BBB) refers to special properties found in the blood vessels of the central nervous system (CNS). The BBB regulates the movement of substances between the blood and brain through tight junctions and transporters (Daneman and Prat, 2015), which prevents toxic compounds from entering the CNS (Bal-Price et al., 2017). Diverse *in silico* models have been developed to predict BBB permeation in the early phase of drug discovery (Liu et al., 2014; Bagchi et al., 2019; Miao et al., 2019). However, the developed models need to be evaluated to determine whether BBB penetration can be reliably predicted for BASs due to differences in chemical space between BASs and pharmaceuticals. To the best of our knowledge, only one study has developed *in silico* models for BASs to predict toxicities toward *Daphnia magna* and fish (Khan

et al., 2019).

Prediction models for inorganic compounds were rarely developed for the endpoint of interest by a regulatory body. Till date, only two models have been developed to predict physicochemical properties (Shin, 2020) and toxicities of inorganic compounds (Toropova et al., 2011) presumably due to lack of data and limited cheminformatics tools for inorganic compounds. A majority of the models have been developed for organic compounds. Given that several hazardous effects of diverse inorganic compounds have been reported, the corresponding *in silico* models need to be developed, particularly to predict endpoints relevant to human health (Yorijufi, 2012; Egorova and Ananikov, 2017).

In this study, we developed BBB permeation prediction models for organic and inorganic BASs. Models were separately developed due to difference in chemical space between organic and inorganic compounds. As the collected BBB permeation data were not BASs for both organic and inorganic compounds, BBB permeation datasets were curated based on structural similarity with BASs in order to develop a model that focused on the chemical space of BASs. For organic BASs, the support vector regression (SVR) model developed with VEGA and quantum mechanical (QM) descriptors achieved the best prediction accuracy ($R^2 = 0.64$). For inorganic BASs, the multiple linear regression (MLR) model developed with logS (water solubility) and boiling point (T_{BP}) descriptors showed the highest prediction accuracy ($R^2 = 0.51$). Applicability domain (AD) analysis presented organic and inorganic BASs reliably predicted by the models. In this study, Quantitative Structure-Activity Relationship (QSAR) validation principles of the Organization for Economic Cooperation and Development (OECD) (OECD, 2006) was followed in model development.

2. Material and methods

2.1. Collection of biocide list

A list of BASs was prepared by combining three different lists: ECHA (ECHA, 2014), Databook of biocides (Wypych and Wypych, 2015), and BASs used in South Korea, which was obtained from a report titled "Development of risk assessment method of biocidal active substances", provided by the Ministry of Environment of Korea. All the BPs in the report are classified according to the regulation on household chemical products and biocidal products by the Ministry of Environment of Korea. Each list of BASs was integrated, and duplicated structures were removed. Salt was removed from the molecular structure. The lists of organic (Table S1) and inorganic (Table S2) BASs are available in the

supplementary information.

2.2. Blood-brain barrier penetration data collection

Quantitative BBB penetration data were collected from ten published articles (Winkler and Burden, 2004; Ma et al., 2005; Konovalov et al., 2007; Zhang et al., 2008; Sá et al., 2010; Yan et al., 2013; Carpenter et al., 2014; Bujak et al., 2015; Wang et al., 2015; Toropov et al., 2017). From the dataset, log-scaled blood brain barrier penetration (logBB) was calculated as follows:

$$\log BB = \log \left(\frac{C_{\text{brain}}}{C_{\text{blood}}} \right)$$

where, C_{brain} and C_{blood} are concentrations of the substances in the brain and blood, respectively. logBB is zero if substance concentrations between the brain and blood are equal. logBB has a positive value if the concentration of substances is higher in the brain than in the blood, and a negative value if the concentration is higher in blood. Duplicated compounds were checked across datasets. When multiple logBB values were obtained for one molecular structure, logBB values were averaged if the difference between maximum and minimum logBB values was equal to or less than 0.5. Otherwise, the structure was discarded from the logBB dataset. Salt was removed from organic molecules to calculate descriptors from active ingredients of organic BASs. For inorganic compounds, only 12 logBB data points were collected.

2.3. Chemical space examination between biocide and logBB dataset

The logBB_[org] (logBB data for organic compounds) was composed of pharmaceuticals. Thus, the chemical space of organic BASs and logBB_[org] were compared in order to select pharmaceuticals that were significantly similar to organic BASs, based on three steps. The first step was based on molecular weight (MW) and Crippen logP, implemented in RDKit (Landrum, 2006), to select compounds in the logBB_[org] similar to organic BASs in terms of size and lipophilicity. Second, the molecular access system fingerprint (MACCS FP) was used to calculate Tanimoto coefficients, and compounds in the logBB_[org] whose Tanimoto coefficients were higher than 0.8 to any of the organic BASs were selected. Third, scaffolds were extracted from organic BASs and compounds in the logBB_[org] using the Murcko scaffold module in RDKit, and pharmaceuticals containing scaffolds that do not exist among organic BASs were removed from the dataset.

The chemical space of inorganic BASs and compounds in the logBB_[inorg] (logBB data for inorganic compounds) was presented with MW and logS, calculated using the model proposed in Shin (2020) in order to examine the similarity between inorganic BASs and compounds in the logBB_[inorg].

2.4. Descriptor calculation and normalization

In the model for logBB_[org] (model_[org]), descriptors were calculated using RDKit, PaDEL (Yap, 2011), VEGA (Benfenati et al., 2019), and Molecular Orbital Package (MOPAC) (Stewart, 2016). RDKit was used to calculate fingerprints (FPs) such as atompair FP, Avalon FP, Avalon count FP, estate FP, layer FP, MACCS FP, Morgan FP, and torsion FP. Morgan FP was calculated from level 2 to 4.2D molecular descriptors implemented in RDKit were also calculated, such as rdMolDescriptors, Descriptors, Crippen, MolSurf, GraphDescriptors, and Estate VSA. The PaDEL-Descriptor GUI version and VEGA were used to calculate 2D descriptors from mol files. QM descriptors were calculated from MOPAC based on the PM7 semi-empirical

method (Stewart, 2013). Before QM calculation, the geometries of all molecules were initially optimized by the Merck molecular force field (MMFF) implemented in RDKit (Tosco et al., 2014), and the molecular structures were subsequently optimized using PM7 methods in MOPAC. QM descriptors were then calculated from the molecular structures converged after QM-based geometry optimization. The calculated QM descriptors were heat of formation, molecular orbital energies, superdelocalizabilities, polarizabilities, and descriptors calculated from the final density matrix of each molecule according to Fukui frontier molecular orbital theory.

In the model for logBB_[inorg] (model_[inorg]), QM descriptors could not be calculated for inorganic compounds present in the gaseous state. Instead, inorganic compounds' physicochemical property calculation models were used (Shin, 2020) to prepare descriptors for inorganic compounds. These models were developed with artificial neural network (ANN) to predict melting point (T_{MP}), T_{BP} , logS, and pyrolysis point (T_{PP}) based on the composition of inorganic compounds. As the ANN models for inorganic physicochemical descriptors were developed from inorganic compounds in solid, liquid, and gaseous states, the four descriptors were calculated for the entire compounds in the logBB_[inorg]. In development of model_[inorg], features were expanded by 1) the four physicochemical properties and MW raised to the power of 2, 3, and 4, 2) exponentiation with the natural number as the base and the physicochemical properties and MW as the exponent, 3) the inverse of physicochemical properties and MW. The prepared descriptors were normalized before model development.

$$\text{Normalized descriptors} = \frac{\text{Descriptors} - \min(\text{descriptors})}{\max(\text{descriptors}) - \min(\text{descriptors})}$$

where min(descriptors) and max(descriptors) are the minimum and maximum values of each descriptor, respectively.

2.5. Data set division and feature selection

Training and external test sets were prepared in order to validate the robustness of the model_[org]. For logBB_[org], one-third of the data was randomly selected as an external test set (Training set: 153 and external test set: 76). The chemical space of the training and external test sets showed that the structural diversity between the training and test sets was similar (Fig. S1). As the number of training sets was 153, the maximum number of descriptors used in the model was limited to 30, according to the ratio of 1–5 between the number of descriptors and data points (Cherkasov et al., 2014). Recursive feature elimination was used in feature selection with SVR and random forest regression (RFR). In the case of model_[inorg] development, a small number of compounds were available for model training. According to the ratio above, only two descriptors were allowed in model development; therefore, all possible combination of descriptors were examined in feature selection.

2.6. Model development and validation

Prediction models were developed using scikit-learn (Pedregosa et al., 2011). OECD validation guideline requires to perform two types of validation: internal and external validations. Therefore, 3-fold cross validation (CV) was performed for internal validation, and external validation was performed with external test set.

For logBB_[org], SVR and RFR models_[org] developed with different descriptors were compared in order to select the best performing descriptors and best machine learning algorithms for the dataset through R^2 and root mean squared error (RMSE) measured from 3-fold CV. In case of logBB_[inorg], available data was small to apply

internal and external validation together. Therefore, MLR model was developed with internal validation alone since MLR is over-fitting resistant. R^2 and RMSE were measured from leave-one-out (LOO).

$$R^2 = \frac{\sum (y_{pred} - y_{exp})^2}{\sum (y_{exp} - \bar{y})^2}$$

$$RMSE = \sqrt{\frac{\sum (y_{pred} - y_{exp})^2}{n}}$$

where, y_{pred} , y_{exp} , and $\bar{y}$ are the predicted, experimental, and averaged experimental values, respectively. n is the number of data points. R^2 was used to evaluate the goodness-of-fit of the model and RMSE was used to check numerical errors. After selecting the descriptor and machine learning algorithm, the number of features and hyperparameters were optimized based on the prediction accuracy of the external test set. The RMSE of the predicted values tends to vary according to the range of experimental values in the dataset. Therefore, the RMSE over the entire range of the endpoint was examined for a fair presentation of model prediction accuracy.

2.7. Y-randomization test

The Y-randomization test aims to test if a model is established by mere coincidence by comparing the models' prediction accuracy from original data (original model) and endpoint-randomized data (random model). In this study, endpoint randomization was conducted 10 times to generate 10 random models. The Z-score was calculated with R^2 as follows:

$$Z = \frac{R^2_{ori} - \mu_{ran}}{\sigma_{ran}}$$

where R^2_{ori} is the R^2 of the original model, and μ_{ran} and σ_{ran} are the mean and standard deviation of R^2 of the random models, respectively. If the Z-score exceeded three, the original model was considered to have established a statistically valid correlation between the descriptors and the endpoint.

2.8. Applicability domain analysis

The William plot (Netzeva et al., 2005) was used to examine the AD of the model. Given that the datasets used to develop models were not exactly BASs, it was crucial to analyze the AD of the models in order to corroborate that the models can make accurate predictions on BASs. The leverage value of the i -th compound (h_i) was calculated as follows:

$$h_i = x_i (X^T X)^{-1} x_i^T$$

where X is a descriptor matrix composed of n chemicals with m descriptors, and x_i is a descriptor vector of the i -th compound with m descriptors. The compounds that exceeded warning leverage (h^*) were considered as out-of-domain of the model, which was calculated as

$$h^* = \frac{3(p+1)}{n}$$

where p is the number of descriptors in the model and n is the

number of compounds in training set.

3. Results and discussion

3.1. Comparison of biocidal active substances' lists from different references

Fig. 1 presents the analysis of the BASs from the three different datasets. The numbers of organic and inorganic BASs in each dataset were 157 and 47 in the ECHA list, 147 and 45 in Databook of biocides, and 150 and 65 in the list from the Ministry of Environment of Korea, respectively (Fig. 1A). In terms of ratio, 23% of BASs in the ECHA list and Databook of biocides belong to inorganic BASs, and 30% of these BASs are being used in Korea. After integration, 257 organic and 95 inorganic BASs were obtained from the three lists by removing duplicates. The proportion of inorganic BASs in the total list of BASs was 27%, which motivated us to develop the model_[inorg] as the ratio of inorganic compounds was relatively high among the integrated BAS lists.

The Venn diagram showed that 75 BASs were common among all three datasets, and a particularly high number of BASs overlapped between the ECHA list and Databook of biocides (167 BASs). The list from the Ministry of Environment of Korea contained 124 BASs that were found in neither the ECHA list nor Databook of biocides (Fig. 1B). A similar pattern was observed in both organic (Fig. 1C) and inorganic (Fig. 1D) BASs.

3.2. Chemical space comparison between logBB data and BASs (organic and inorganic)

The chemical space of compounds in the logBB_[org] and organic BASs showed significant similarity when MW was less than 520 Da and greater than 15, and logP was less than 6.2 and greater than -3. According to the scaffold structure analysis, multiple scaffolds that were absent in organic BASs were found within compounds in the logBB_[org] (Fig. 2). Fig. 2A mainly shows the 7-membered rings, and Fig. 2B shows multiple fused rings. Scaffolds in Fig. 2 were found among logBB_[org] even after curation based on Tanimoto coefficients. Thus, organic compounds containing the scaffolds were manually removed from logBB_[org]. After curation, the structural similarity between logBB_[org] and organic BASs was improved (Fig. 3A), and the logBB_[org] values ranged from -1.88 to 1.38 (Table S3).

Examination of chemical space led to removal of Rn from logBB_[inorg] as no inorganic BASs were similar to Rn in the chemical space. Inorganic BASs whose MW was lower than 170 and greater than 15 whose logS was lower than 1.5 and greater than -2.2 were considered to be similar to the logBB_[inorg] (Fig. 3C), and the logBB_[inorg] values ranged between -0.16 and 0.6 (Table S4). Winkler and Burden developed logBB prediction model with dataset including organic and inorganic compounds together using property-based descriptors, topological indices, and eigenvalues of modified adjacency matrices (Winkler and Burden, 2004). Therefore, Winkler and Burden's model can be applied to organic and inorganic molecules as long as the descriptors can be calculated from an input molecule. In order to consider the possibility of general model development for both organic and inorganic BASs, molecular structures of organic and inorganic BASs were examined in this study. In case of BASs, majority of inorganic BASs contained metal atoms in the structures, which made it impossible to calculate common descriptors between organic and inorganic BASs with currently available cheminformatics tools. Therefore, we concluded that specific model targeting chemical space of inorganic BASs should be developed.

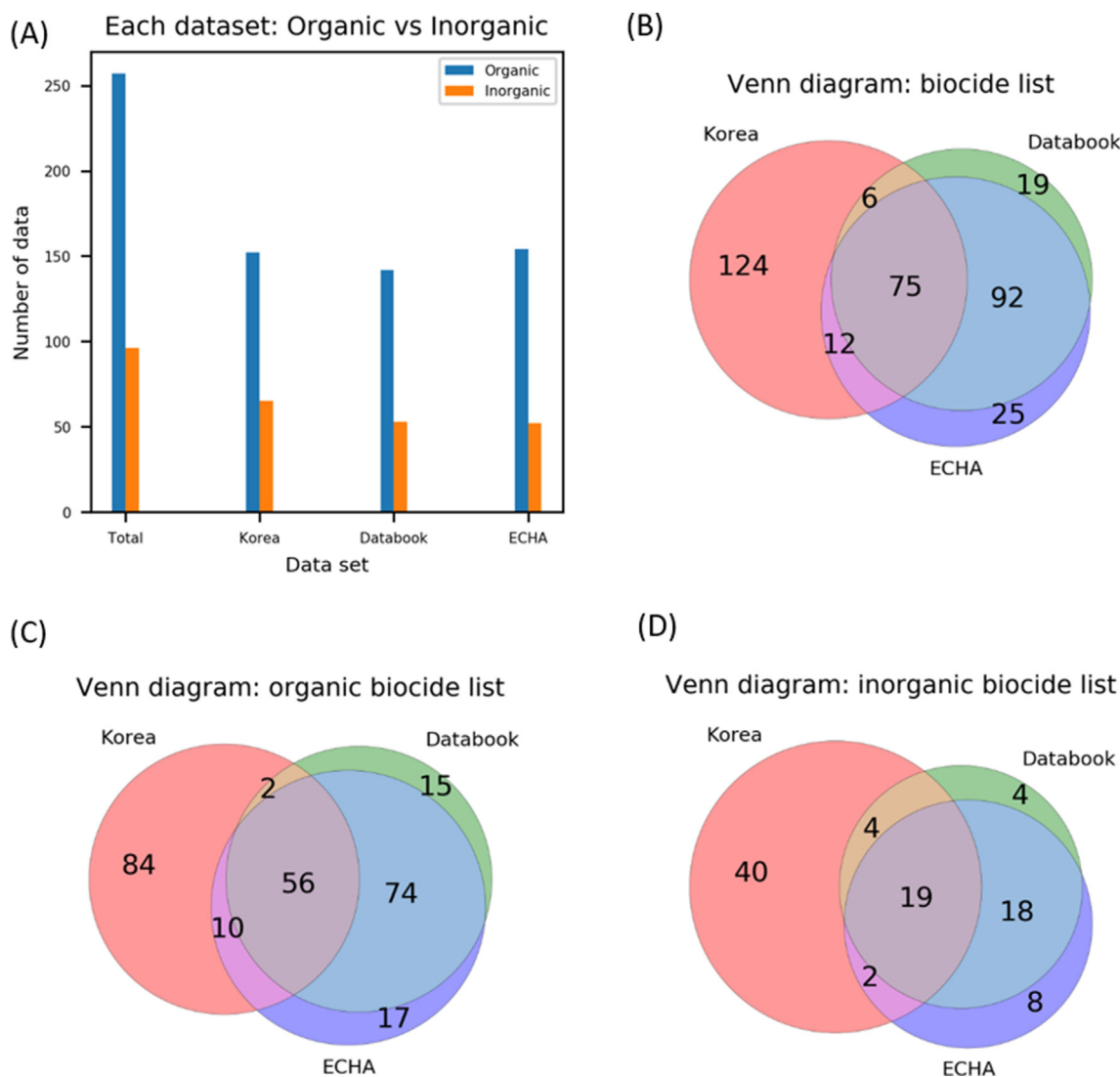


Fig. 1. Biocidal active substances between three different lists were analyzed. BAS list from Korea was obtained from the Ministry of Environment of Korea. (A) A comparison between the number of organic and inorganic BASs. Venn diagrams for (B) all BASs, (C) organic BASs, and (D) inorganic BASs.

3.3. Development and validation of logBB prediction model for organic BASs

The prediction accuracy of the RFR models_[org] on the training set was higher than that of the SVR models_[org], whereas the prediction accuracy in 3-fold CV was similar between the RFR and SVR models_[org] on average (Table 1). Two conditions were considered in the model selection: 1) R^2 of training and test sets should not deviate by more than 0.3 (Eriksson et al., 2003; Epa et al., 2012), and 2) R^2 from internal validation should exceed 0.6 (Alexander et al., 2015). Only one SVR model_[org] developed with VEGA and QM descriptors met these conditions. Therefore, further hyperparameter optimization was conducted in the SVR model_[org] with VEGA and QM descriptors. The highest accuracy on the external test set was obtained with the SVR model_[org] when 30 descriptors were used, the radial basis function was a kernel function, C was 30, and epsilon (ϵ) was 0.01 (Table 2). As R^2 of the test set exceeded 0.6, the predicted and observed values showed a linear distribution close to the best fit line (Fig. 3B). The RMSE of the test set was 0.37, which was a relatively small numerical error compared to the entire range of logBB in the test set (12.42%). As the best model_[org] also passed the Y-randomization test (Z-score: 3.82), we concluded that the

SVR model_[org] can be used in the quantitative prediction of logBB.

Table S5 presents a list of descriptors selected in the model_[org] with Pearson correlation coefficient between selected descriptors and logBB_[org]. The model_[org] was developed with 25 VEGA descriptors and 5 QM descriptors. The correlation analysis showed that MlogP showed the highest correlation with logBB (0.44). logP was commonly selected in many logBB prediction models (Bujak et al., 2015) as the lipophilicity of the compounds significantly affects the passive transportation of chemicals through the BBB. Even though logP is highly correlated with logBB, it is still challenging to accurately predict logBB due to the diverse transporters that are involved in BBB permeation. As the conformational changes in transporters are dynamic, a wide range of molecules with different sizes and properties interact with transporters. Presumably, other descriptors, except MlogP, were related to interactions between BASs and transporters expressed on the BBB.

3.4. Prediction accuracy comparison between developed model for organic BASs and an available model for pharmaceuticals

Even though the model_[org] was developed with a carefully curated dataset, further validation was required to corroborate that

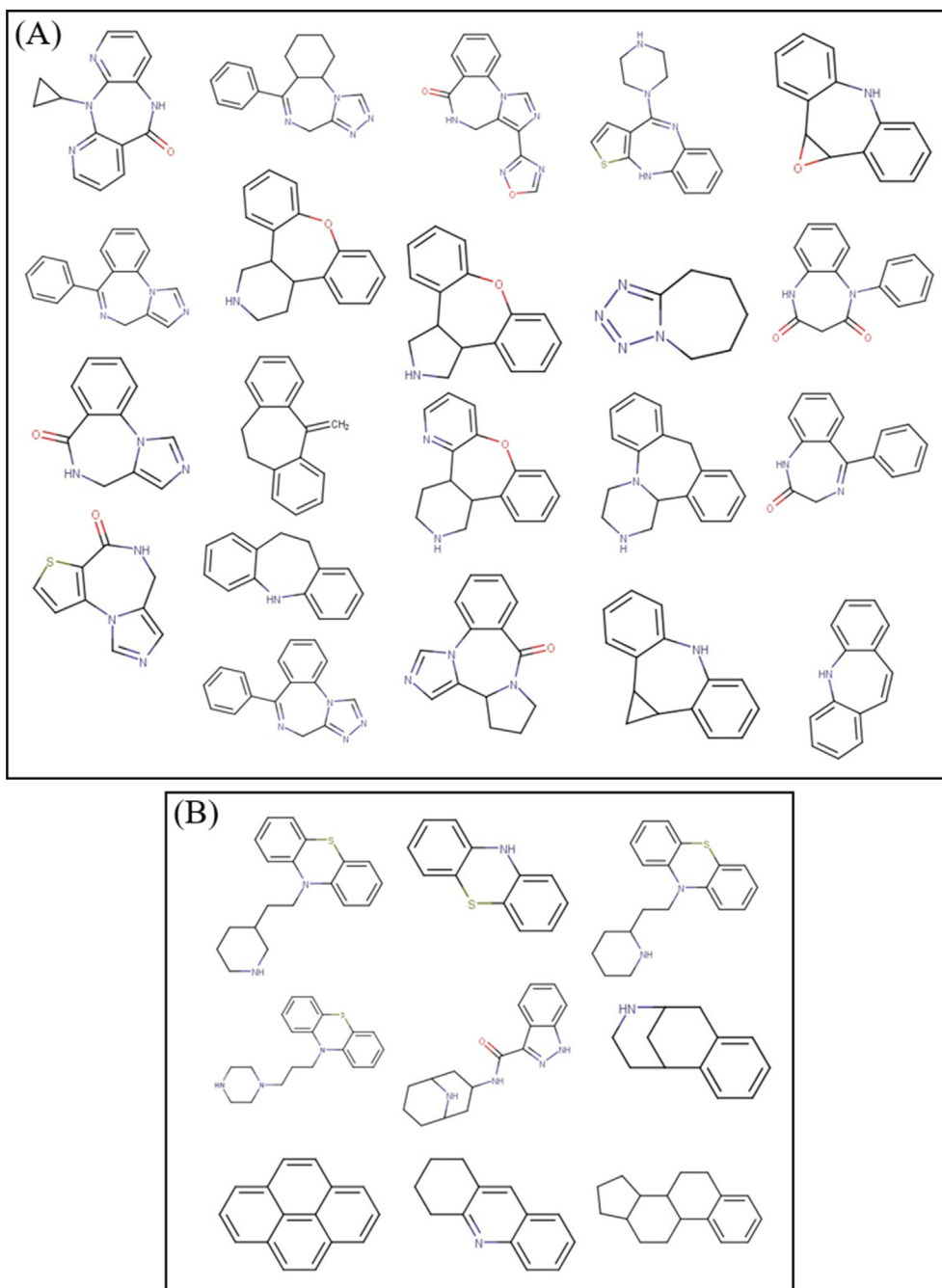


Fig. 2. Structures of Murcko scaffolds absent among organic BASs were extracted from $\log\text{BB}_{[\text{org}]}$. (A) 7-membered ring structures were not found in any organic BASs. (B) Pharmaceuticals tend to have more number of rings in their molecular structure compared to that of organic BASs.

the prediction accuracy of the $\text{model}_{[\text{org}]}$ on organic BASs was significantly improved against $\log\text{BB}$ prediction models for pharmaceuticals as $\log\text{BB}_{[\text{org}]}$ was basically composed of pharmaceutical compounds. Therefore, the prediction accuracy of the available quantitative $\log\text{BB}$ prediction model for pharmaceuticals was examined on organic BASs. All freely available BBB penetration prediction models were classification models, except the one implemented in PreADMET (<https://preadmet.bmdrc.kr/>). The $\log\text{BB}$ model in PreADMET showed higher RMSE for organic BASs compared to the $\text{model}_{[\text{org}]}$ in this study, as the RMSE on the training and test sets of $\log\text{BB}_{[\text{org}]}$ was 2.99 and 3.84, respectively. In the prediction results, the $\log\text{BB}$ model in PreADMET tended to

overestimate $\log\text{BB}$ of organic BASs, which could be due to the main goal of the model: searching drug candidates with good BBB penetration properties. This comparison showed that the $\log\text{BB}$ prediction model for pharmaceuticals was inappropriate for screening BBB-permeable BASs. Thus we concluded that the $\text{model}_{[\text{org}]}$ in this study made significant prediction accuracy improvement towards organic BASs.

3.5. Development and validation of $\log\text{BB}$ prediction model for inorganic BASs

Table 3 shows the R^2 and RMSE values for $\log\text{BB}_{[\text{inorg}]}$ with

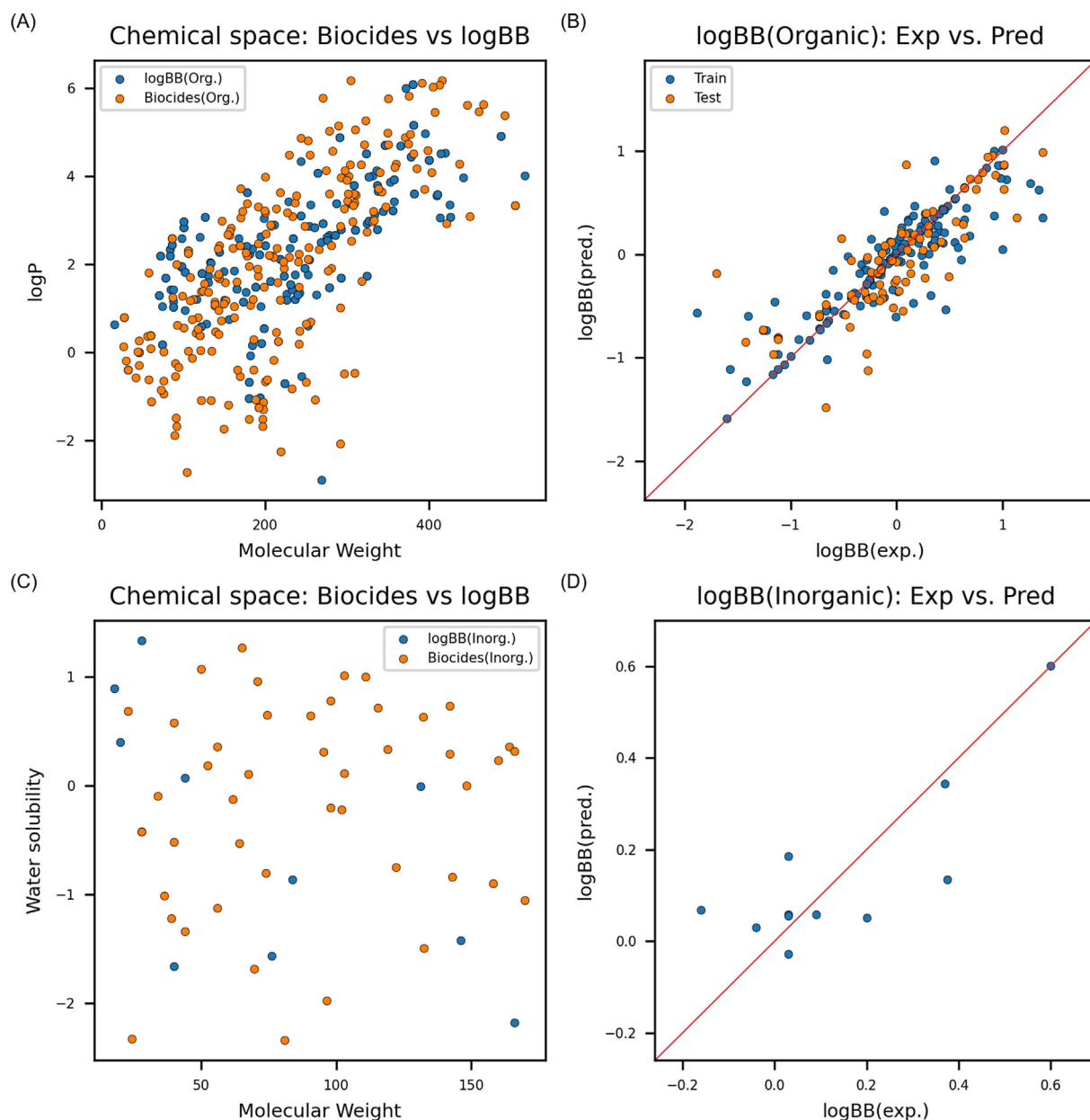


Fig. 3. Chemical space and goodness-of-fit. (A) Comparison of chemical space between $\log BB_{[org]}$ and organic BASs. (B) Predicted and observed $\log BB_{[org]}$, plotted along with the best fit line (red). (C) Comparison of chemical space between $\log BB_{[inorg]}$, and inorganic BASs. (D) Predicted and observed $\log BB_{[inorg]}$, plotted along with the best fit line (red). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

different combinations of descriptors. According to prediction accuracy on training set, the highest R^2 was achieved when the MLR model_[inorg] was developed with MW^4 and $1/T_{BP}$, which achieved only R^2 0.2 in LOO. In LOO, highest R^2 was achieved in MLR model_[inorg] with $\log S^3$ and $1/T_{BP}$, $R^2(\text{train})$ 0.66 & $R^2(\text{LOO})$ 0.51.

$$\log BB_{inorg} = -0.36 \log S^3 - 0.45 \frac{1}{T_{BP}} + 0.78$$

Table 2 shows the model performance, and the model also passed the Y-randomization test (Z-score: 3.74). Fig. 3D shows the distribution of observed and predicted values along with the best fit line.

$\log S$ and $\log P$ were highly correlated. Given that a negative value was assigned for $\log S$ in the model_[inorg], low water solubility (or high lipophilicity) was also significant in the prediction of BBB

permeation for inorganic compounds. In the model_[inorg], a negative coefficient was also assigned to an inverse of T_{BP} . The size of a molecule is another significant property to predict BBB permeation, and T_{BP} is a descriptor related with size of molecule since it is affected by intermolecular forces. To the best of our knowledge, the model_[inorg] is the first model to predict $\log BB$ for inorganic compounds. The model achieved good prediction accuracy with a linear model based on two descriptors that were significant in BBB permeation. Therefore, we concluded that the model_[inorg] can be used for prediction of $\log BB$ for inorganic compounds within AD.

3.6. Applicability domain analysis of the models for organic and inorganic BASs

The AD of model_[org] and model_[inorg] was analyzed in order to

Table 1
3-fold cross-validation results for logBB prediction models targeting organic BASs.

Descriptors	Support Vector Regressor (SVR)				Random Forest Regressor (RFR)			
	R ² (tr)	R ² (cv)	RMSE (tr)	RMSE (cv)	R ² (tr)	R ² (cv)	RMSE (tr)	RMSE (cv)
Atompair FP	0.82	0.42 ± 0.04	0.26	0.46 ± 0.03	0.9	0.52 ± 0.06	0.19	0.42 ± 0.02
Avalon FP	0.74	0.53 ± 0.07	0.32	0.42 ± 0.05	0.89	0.53 ± 0.12	0.2	0.42 ± 0.06
Avalon count FP	0.98	0.45 ± 0.06	0.08	0.45 ± 0.07	0.89	0.50 ± 0.05	0.2	0.43 ± 0.06
Estate FP	0.98	0.45 ± 0.03	0.09	0.45 ± 0.05	0.91	0.26 ± 0.06	0.18	0.53 ± 0.08
Layer FP	0.77	0.45 ± 0.04	0.29	0.46 ± 0.04	0.86	0.21 ± 0.19	0.23	0.51 ± 0.06
MACCS FP	0.63	0.48 ± 0.02	0.37	0.44 ± 0.03	0.87	0.49 ± 0.11	0.22	0.43 ± 0.07
Morgan FP (level 2)	0.38	0.12 ± 0.03	0.48	0.57 ± 0.04	0.85	0.51 ± 0.04	0.2	0.43 ± 0.03
Morgan FP (level 3)	0.5	0.32 ± 0.01	0.44	0.50 ± 0.02	0.86	0.33 ± 0.08	0.23	0.50 ± 0.01
Morgan FP (level 4)	0.43	0.29 ± 0.09	0.46	0.51 ± 0.06	0.81	0.42 ± 0.05	0.27	0.47 ± 0.04
Autocorrelation 2D	0.55	0.42 ± 0.13	0.41	0.46 ± 0.07	0.91	0.44 ± 0.11	0.18	0.46 ± 0.07
Torsion FP	0.4	0.25 ± 0.05	0.48	0.53 ± 0.06	0.81	0.28 ± 0.02	0.27	0.52 ± 0.01
PaDEL 2D	0.83	0.51 ± 0.07	0.25	0.43 ± 0.06	0.90	0.50 ± 0.13	0.19	0.43 ± 0.03
RDKit 2D	0.65	0.56 ± 0.01	0.37	0.41 ± 0.05	0.92	0.57 ± 0.07	0.18	0.40 ± 0.05
QM	0.51	0.39 ± 0.09	0.43	0.48 ± 0.08	0.88	0.28 ± 0.08	0.22	0.52 ± 0.03
Vega	0.86	0.56 ± 0.03	0.23	0.41 ± 0.01	0.93	0.49 ± 0.11	0.17	0.44 ± 0.06
RDKit 2D + QM	0.7	0.58 ± 0.06	0.34	0.40 ± 0.04	0.92	0.52 ± 0.12	0.17	0.42 ± 0.06
VEGA + QM	0.74	0.60 ± 0.04	0.31	0.39 ± 0.06	0.92	0.42 ± 0.16	0.18	0.45 ± 0.08

*tr: training set.

Table 2
Best models for organic and inorganic BASs.

BAS type	Model parameters	Training set			Y-ran.	Validation		
		R ²	RMSE			R ²	RMSE	
Organic	SVR (C:30, ϵ : 0.01, RBF)	0.76	0.3	9.20%	3.82	0.64	0.37	12.42%
Inorganic	MLR	0.66	0.12	15.79%	3.74	0.51	0.15	19.74%

*Y-ran.: Z-score from Y-randomization test.

Table 3
Prediction performances of the logBB prediction models targeting inorganic BASs.

Descriptors	Training set			Leave-one-out		
	R ²	RMSE		R ²	RMSE	
logS, T _{PP}	0.68	0.12	15.79%	0.41	0.16	21.05%
logS, T _{BP}	0.68	0.12	15.79%	0.42	0.16	21.05%
logS, T _{BP}	0.67	0.12	15.79%	0.31	0.18	23.68%
T _{BP} , T _{MP} ³	0.72	0.11	14.47%	0.4	0.16	21.05%
T _{PP} , logS ²	0.53	0.14	18.42%	0.22	0.19	25.00%
T _{PP} , e ^{-logS}	0.65	0.12	15.79%	0.29	0.18	23.68%
MW ³ , 1/T _{BP}	0.77	0.1	13.16%	0.34	0.17	22.37%
logS³, 1/T_{BP}	0.66	0.12	15.79%	0.51	0.15	19.74%
T _{MP} ³ , T _{MP} ⁴	0.48	0.15	19.74%	0.28	0.18	23.68%
MW ⁴ , 1/T _{BP}	0.79	0.1	13.16%	0.2	0.19	25.00%

Table 4
Applicability domain analysis for logBB prediction models targeting organic and inorganic BASs.

Organic BASs (h*: 0.61)		Applicability domain		Ratio (In-Domain/Total)	
Train set		In-Domain	Out-Domain		
Test set		149	4		97%
Total biocides		72	6		92%
Biocides within chemical space (15 < MW < 520, -3 < logP < 6.2, -2.28 < logBB < 1.78)		199	63		76%
Inorganic BASs (h*: 0.82)					81%
		Applicability domain		Ratio (In-Domain/Total)	
Data set		In-Domain	Out-Domain		
Total biocides		11	0		100%
Biocides within chemical space (15 < MW < 170, -2.2 < logS < 1.5, -0.36 < logBB < 0.8)		55	26		68%
		43	0		100%

difference was found in CATS2D_9_DA descriptor, which is related with hydrogen donor and acceptor. Additionally three more descriptors related with hydrogen bonding were listed in top10 descriptor, two more CATS2D descriptors and the number of carboxylic acids (nRCOOH). As a result, averaged MLogP for BASs out-of-domain were lower than BASs in-domain. Therefore, AD analysis revealed that organic BASs out of AD tended to be hydrophilic compared to that within AD due to relatively higher number of hydrogen bond donor and acceptor functional groups, which can be related with the fact that the training set for model_[org] was composed of drug-like compounds. Even though molecules highly similar with organic BASs were carefully selected, the training set was still relatively lipophilic compared to organic BASs.

Model_[inorg] was developed with inorganic compounds whose MW was lower than 170 and higher than 15, and whose logS was greater than −2.2 and lower than 1.5. logBB_[inorg] values ranged between −0.16 and 0.6, and reliable predicted logBB was considered within −0.36 and 0.8 by considering RMSE of the model_[inorg]. Among 81 inorganic BASs, 43 compounds (53%) were within the range of MW, logS, and logBB, and all inorganic BASs within chemical space were within AD of the model_[inorg] (Table S2). In descriptor space analysis (Fig. S3), most of inorganic BASs out of chemical space had high T_{BP} or low logS. As cube of logS and inverse of T_{BP} were selected in the model, slight change of logS and T_{BP} made dramatic variation in prediction outcome. Therefore, reliable prediction results were guaranteed only if inorganic BASs were within AD.

4. Conclusions

In this study, we developed quantitative BBB penetration prediction models for organic and inorganic BASs. Currently, BBB data for BASs are not available. Therefore, the chemical space of the collected datasets was carefully examined, and data points satisfying structural similarity criteria were selected in the model development. The developed model_[org] achieved good prediction accuracy in external validation ($R^2 = 0.64$) and outperformed the logBB prediction model for pharmaceuticals. The model_[inorg] is the first model for the prediction of BBB permeation for inorganic compounds, and it was developed with good prediction accuracy in internal validation ($R^2 = 0.51$) based on logS³ and 1/T_{BP}. AD analysis was conducted with leverage methods in order to make reliable predictions from the models. The models can be used to predict BBB permeation of currently available BASs and to support the safe-by-design concept in BAS development by screening BBB-permeable BASs using chemical structure alone.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by Korea Environmental Industry & Technology Institute (KEITI) through Technology Program for establishing biocide safety management, funded by Korea Ministry of Environment (MOE) (2019002490006), by National Research Foundation of Korea (NRF) funded by the Korean government (MSIT) (NRF-2019R1F1A1061955), and by Korea Institute of Toxicology (KIT), Republic of Korea (1711121217).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2021.130330>.

Credit author statement

Hyun Kil Shin: Methodology, Software, Writing – original draft, Visualization, Sangwoo Lee: Project administration, Resources, Formal analysis, Donggon Yoo: Investigation, Data curation Seun Min Park: Investigation, Data curation Ha-Na Oh: Resources, Validation, Woo-Keun Kim: Supervision, Funding acquisition, Writing – review & editing, Myung-Gyun Kang: Conceptualization, Writing – review & editing

References

- Alexander, D.L.J., Tropsha, A., Winkler, D.A., 2015. Beware of R^2 : simple, unambiguous assessment of the prediction accuracy of QSAR and QSPR models. *J. Chem. Inf. Model.* 55, 1316–1322.
- Bagchi, S., Chhibber, T., Lahooti, B., Verma, A., Borse, V., Jayant, R.D., 2019. *in-vitro* blood-brain barrier models for drug screening and permeation studies: an overview. *Drug Des. Dev. Ther.* 13, 3591–3605.
- Bal-Price, A., Lein, P.J., Keil, K.P., Sethi, S., Shafer, T., Barenys, M., Fritsche, E., Sachana, M., Meek, M.E., 2017. Developing and applying the adverse outcome pathway concept for understanding and predicting neurotoxicity. *Neurotoxicology* 59, 240–255.
- Benfenati, E., Roncaglioni, A., Lombardo, A., Manganaro, A., 2019. Integrating QSAR, read-across, and screening tools: the VEGA HUB platform as an example. In: Hong, H. (Ed.), *Advances in Computational Toxicology: Methodologies and Applications in Regulatory Science*. Springer International Publishing, Switzerland, pp. 365–381.
- Bujak, R., Struck-Lewicka, W., Kaliszan, M., Kaliszan, R., Markuszewski, M.J., 2015. Blood–brain barrier permeability mechanisms in view of quantitative structure–activity relationships (QSAR). *J. Pharmaceut. Biomed.* 108, 29–37.
- Burnett, C.L., Bergfeld, W.F., Belsito, D.V., Klaassen, C.D., Marks, J.G., Shank, R.C., Slaga, T.J., Snyder, P.W., Andersen, F.A., 2010. Final report of the safety assessment of methylisothiazolinone. *Int. J. Toxicol.* 29, 1875–2135.
- Carpenter, Timothy S., Kirshner, Daniel A., Lau, Edmond Y., Wong, Sergio E., Nilmeier, Jerome P., Lightstone, Felice C., 2014. A method to predict blood-brain barrier permeability of drug-like compounds using molecular dynamics simulations. *Biophys. J.* 107, 630–641.
- Cherkasov, A., Muratov, E.N., Fourches, D., Varnek, A., Baskin, I.I., Cronin, M., Dearden, J., Gramatica, P., Martin, Y.C., Todeschini, R., Consonni, V., Kuz'min, V.E., Cramer, R., Benigni, R., Yang, C., Rathman, J., Terfloth, L., Gasteiger, J., Richard, A., Tropsha, A., 2014. QSAR modeling: where have you been? where are you going to? *J. Med. Chem.* 57, 4977–5010.
- Christen, V., Rusconi, M., Crettaz, P., Fent, K., 2017. Developmental neurotoxicity of different pesticides in PC-12 cells *in vitro*. *Toxicol. Appl. Pharmacol.* 325, 25–36.
- Collier, P.J., Ramsey, A., Waigh, R.D., Douglas, K.T., Austin, P., Gilbert, P., 1990. Chemical reactivity of some isothiazolone biocides. *J. Appl. Bacteriol.* 69, 578–584.
- Costa, L.G., 2019. Toxic effects of pesticides. In: Klaassen, C.D. (Ed.), *Casarett & Doull's Toxicology: the Basic Science of Poisons*, ninth ed. McGraw-Hill Education, New York, pp. 1061–1069.
- Daneman, R., Prat, A., 2015. The blood-brain barrier. *Cold Spring Harb. Perspect. Biol.* 7, a020412–a020412.
- Du, S., McLaughlin, B., Pal, S., Aizenman, E., 2002. *In vitro* neurotoxicity of methylisothiazolinone, a commonly used industrial and household biocide, proceeds via a zinc and extracellular signal-regulated kinase mitogen-activated protein kinase-dependent pathway. *J. Neurosci.* 22, 7408.
- ECHA, 2014. List of Active Substances and Suppliers.
- ECHA, 2016. Practical Guide on Biocidal Products Regulation. European chemicals agency.
- Egorova, K.S., Ananikov, V.P., 2017. Toxicity of metal compounds: knowledge and myths. *Organometallics* 36, 4071–4090.
- Epa, V.C., Burden, F.R., Tassa, C., Weissleder, R., Shaw, S., Winkler, D.A., 2012. Modeling biological activities of nanoparticles. *Nano Lett.* 12, 5808–5812.
- Eriksson, L., Jaworska, J., Worth Andrew, P., Cronin Mark, T.D., McDowell Robert, M., Gramatica, P., 2003. Methods for reliability and uncertainty assessment and for applicability evaluations of classification- and regression-based QSARs. *Environ. Health Perspect.* 111, 1361–1375.
- Hahn, S., Schneider, K., Gartsier, S., Heger, W., Mangelsdorf, I., 2010. Consumer exposure to biocides-identification of relevant sources and evaluation of possible health effects. *Environ. Health* 9, 7–7.
- He, K., Huang, J., Lagenaur, C.F., Aizenman, E., 2006. Methylisothiazolinone, a neurotoxic biocide, disrupts the association of Src family tyrosine kinases with focal adhesion kinase in developing cortical neurons. *J. Pharmacol. Exp. Ther.* 317, 1320.
- Kahrilas, G.A., Blotvogel, J., Stewart, P.S., Borch, T., 2015. Biocides in hydraulic

- fracturing fluids: a critical review of their usage, mobility, degradation, and toxicity. *Environ. Sci. Technol.* 49, 16–32.
- Khan, K., Khan, P.M., Lavado, G., Valsecchi, C., Pasqualini, J., Baderna, D., Marzo, M., Lombardo, A., Roy, K., Benfenati, E., 2019. QSAR modeling of daphnia magna and fish toxicities of biocides using 2D descriptors. *Chemosphere* 229, 8–17.
- Kononov, D.A., Coomans, D., Deconinck, E., Vander Heyden, Y., 2007. Benchmarking of QSAR Models for blood-brain barrier permeation. *J. Chem. Inf. Model.* 47, 1648–1656.
- Landrum, G., 2006. RDKit: Open-Source Cheminformatics.
- Lee, Y.-S., Nam, M.-W., Kim, H.-I., Kim, H.-C., Mun, E.-C., Park, S.-Y., 2020. Mental health impact on a humidifier disinfectant disaster victim: a case report. *Ann. Occup. Environ. Med.* 32.
- Liu, H., Wang, L., Lv, M., Pei, R., Li, P., Pei, Z., Wang, Y., Su, W., Xie, X.-Q., 2014. AlzPlatform: an Alzheimer's disease domain-specific chemogenomics knowledgebase for polypharmacology and target identification research. *J. Chem. Inf. Model.* 54, 1050–1060.
- London, L., Beseler, C., Bouchard, M.F., Bellinger, D.C., Colosio, C., Grandjean, P., Harari, R., Kootbodin, T., Kromhout, H., Little, F., Meijster, T., Moretto, A., Rohlman, D.S., Stallones, L., 2012. Neurobehavioral and neurodevelopmental effects of pesticide exposures. *Neurotoxicology* 33, 887–896.
- Ma, X.-I., Chen, C., Yang, J., 2005. Predictive model of blood-brain barrier penetration of organic compounds. *Acta Pharmacol. Sin.* 26, 500–512.
- Miao, R., Xia, L.-Y., Chen, H.-H., Huang, H.-H., Liang, Y., 2019. Improved classification of blood-brain-barrier drugs using deep learning. *Sci. Rep.* 9, 8802.
- Netzeva, T.I., Worth, A.P., Aldenberg, T., Benigni, R., Cronin, M.T.D., Gramatica, P., Jaworska, J.S., Kahn, S., Klopman, G., Marchant, C.A., Myatt, G., Nikolova-Jeliazkova, N., Patlewicz, G.Y., Perkins, R., Roberts, D.W., Schultz, T.W., Stanton, D.T., van de Sandt, J.J.M., Tong, W., Veith, G., Yang, C., 2005. Current status of methods for defining the applicability domain of (quantitative) structure-activity relationships: the report and recommendations of ECVAM workshop 521.2. *Altern. Lab. Anim.* 33, 155–173.
- OECD, 2006. Guidance Document on the Validation of (Quantitative) Structure-Activity Relationships [(Q)SAR] Models. IOMC (Inter-organization programme for the sound management of chemicals), Paris.
- Pangyi, K., Jong-Han, L., 2016. The humidifier disinfectant scandal: the need for vigorous government oversight of chemicals and household products to secure public safety. *Environ. Health Toxicol.* 31 e2016012–e2016012.
- Park, M.S., Kim, Y.D., Kim, B.-M., Kim, Y.-J., Kim, J.K., Rhee, J.-S., 2016. Effects of antifouling biocides on molecular and biochemical defense system in the gill of the pacific oyster *Crassostrea gigas*. *PloS One* 11, e0168978.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perot, M., Duchesnay, E., 2011. Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.
- Roberts Eric, M., English Paul, B., Grether Judith, K., Windham Gayle, C., Somberg, L., Wolff, C., 2007. Maternal residence near agricultural pesticide applications and autism spectrum disorders among children in the California central valley. *Environ. Health Perspect.* 115, 1482–1489.
- Sá, M.M.d., Pasqualoto, K.F.M., Rangel-Yagui, C.d.O., 2010. A 2D-QSPR approach to predict blood-brain barrier penetration of drugs acting on the central nervous system. *Braz. J. Pharm. Sci.* 46, 741–751.
- Schweer, C., 2010. Biocides-risks and Alternatives. PAN Germany, Pestizid Aktions-Netzwerk, p. 5.
- Shin, H.K., 2020. Electron configuration-based neural network model to predict physicochemical properties of inorganic compounds. *RSC Adv.* 10, 33268–33278.
- Silva, V., Silva, C., Soares, P., Garrido, E.M., Borges, F., Garrido, J., 2020. Isothiazolinone biocides: chemistry, biological, and toxicity profiles. *Molecules* 25, 991.
- Smolka, S., 2013. Biocide-treated Consumer Products. PAN Germany, Pestizid Aktions-Netzwerk, p. 3.
- Sobanska, M.A., Aschberger, K., Rasmussen, K., Pedersen, F., Kappes, D., 2003. The review programme in the European Union for existing biocidal active substances—outcome of the notification process. *Environ. Sci. Pol.* 6, 513–519.
- Soleimani, S.M.A., Ekhtiari, H., Cadet, J.L., 2016. Chapter 2 - drug-induced neurotoxicity in addiction medicine: from prevention to harm reduction. In: Ekhtiari, H., Paulus, M. (Eds.), *Progress in Brain Research*. Elsevier, pp. 19–41.
- Steward, J.J.P., 2016. MOPAC2016. Stewart Computational Chemistry, Colorado Springs, CO, USA.
- Stewart, J.J.P., 2013. Optimization of parameters for semiempirical methods VI: more modifications to the NDDO approximations and re-optimization of parameters. *J. Mol. Model.* 19, 1–32.
- Tong, Z.-B., Hogberg, H., Kuo, D., Sakamuru, S., Xia, M., Smirnova, L., Hartung, T., Gerhold, D., 2017. Characterization of three human cell line models for high-throughput neuronal cytotoxicity screening. *J. Appl. Toxicol.* 37, 167–180.
- Toropov, A.A., Toropova, A.P., Beeg, M., Gobbi, M., Salmons, M., 2017. QSAR model for blood-brain barrier permeation. *J. Pharmacol. Toxicol. Methods* 88, 7–18.
- Toropov, A.P., Toropov, A.A., Benfenati, E., Gini, G., 2011. QSAR modelling toxicity toward rats of inorganic substances by means of CORAL. *Cent. Eur. J. Chem.* 9, 75–85.
- Tosco, P., Stiefl, N., Landrum, G., 2014. Bringing the MMFF force field to the RDKit: implementation and validation. *J. Cheminf.* 6, 37.
- Wagner-Schuman, M., Richardson, J.R., Auinger, P., Braun, J.M., Lanphear, B.P., Epstein, J.N., Yolton, K., Froehlich, T.E., 2015. Association of pyrethroid pesticide exposure with attention-deficit/hyperactivity disorder in a nationally representative sample of U.S. children. *Environ. Health* 14, 44–44.
- Wang, W., Kim, M.T., Sedykh, A., Zhu, H., 2015. Developing enhanced blood-brain barrier permeability models: integrating external bio-assay data in QSAR modeling. *Pharm. Res. (N. Y.)* 32, 3055–3065.
- Winkler, D.A., Burden, F.R., 2004. Modelling blood-brain barrier partitioning using Bayesian neural nets. *J. Mol. Graph. Model.* 22, 499–505.
- Wypych, A., Wypych, G., 2015. 2 - information on data fields. In: Wypych, A., Wypych, G. (Eds.), *Databook of Biocides*. ChemTec Publishing, pp. 11–21.
- Yan, A., Liang, H., Chong, Y., Nie, X., Yu, C., 2013. *In silico* prediction of blood-brain barrier permeability. *SAR QSAR Environ. Res.* 24, 61–74.
- Yap, C.W., 2011. PaDEL-descriptor: an open source software to calculate molecular descriptors and fingerprints. *J. Comput. Chem.* 32, 1466–1474.
- Yorifuji, B., 2012. *Wonderful Life with the Elements*. Kagaku Dojin Publishing Company, Tokyo.
- Zhang, L., Zhu, H., Oprea, T.I., Golbraikh, A., Tropsha, A., 2008. QSAR modeling of the blood-brain barrier permeability for diverse organic compounds. *Pharm. Res. (N. Y.)* 25, 1902.